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Mirror Symmetry of the NMR Spectrum and the Connection with the Structure of Spin Hamiltonian Matrix Representations

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Abstract

This document provides a comprehensive theoretical framework for understanding the symmetry properties of High-Resolution NMR spectra. We analyze the conditions under which a spectrum exhibits mirror symmetry (palindromicity). We demonstrate that such symmetry can arise from two distinct mechanisms: (1) the direct geometric bisymmetry of the Hamiltonian matrix in a generalized canonical basis (typical for balanced systems like A_nB_n or A_nX_n), and (2) a more fundamental property of topological isospectrality (similarity) under parameter exchange induced by the internal symmetry of the spin system, which applies even when the matrix lacks geometric symmetry (as observed in $AA'BB'$ systems).

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1 General Formulas

The Hamiltonian of a nuclear spin system in a strong magnetic field (high-resolution NMR) is represented as the sum of the Zeeman interaction and spin-spin interactions:

$$\hat{H} = \hat{H}_Z + \hat{H}_{SS} = \hat{H}_Z + (\hat{H}_J + \hat{H}_D). \quad (1)$$

Where the components are defined as follows:

- **Zeeman Interaction ($\hat{H}_Z$):** Interaction of spins with the external magnetic field B_0.

$$\hat{H}_Z = \sum_{i=1}^{n} \nu_i \hat{I}_{iz}. \quad (2)$$

  Here ν_i is the resonance frequency of the i-th nucleus, and $\hat{I}_{iz}$ is the operator of the z-component of the spin angular momentum.

- **Spin-Spin Interaction ($\hat{H}_{SS}$):** The sum of isotropic scalar (J) and secular dipolar (D) contributions. In the secular approximation (high field), only the terms commuting with total I_z are retained:

$$\hat{H}_{SS} = \sum_{i<j} J_{ij}(\hat{\vec{I}}_i \cdot \hat{\vec{I}}_j) + \sum_{i<j} d_{ij}(3\hat{I}_{iz}\hat{I}_{jz} - \hat{\vec{I}}_i \cdot \hat{\vec{I}}_j), \quad (3)$$

  where J_{ij} is the scalar coupling constant and d_{ij} is the residual dipolar coupling constant (if present).

2 Canonical Basis and Matrix Symmetries

It is important to distinguish between the symmetry of a specific spin system and the general symmetry properties of the Hamiltonian matrix representation. In this section, we define the **Canonical Basis** – a general-purpose basis applicable to any spin system. We then demonstrate that in this basis, the Hamiltonian matrices possess inherent symmetries (persymmetry and bisymmetry) that stem solely from the algebra of spin operators.

For the matrix representation, we employ the basis of simple products of single-spin functions (*Product Basis*). Each state $|\psi_k\rangle$ of a system of N spins (where $I = 1/2$) is defined by the set of magnetic quantum numbers $m_i = \pm 1/2$:

$$|\psi_k\rangle = |m_1^{(k)}, m_2^{(k)}, \ldots, m_N^{(k)}\rangle. \quad (4)$$

2.1 Basis Ordering and Structure

The basis functions are ordered to satisfy two fundamental conditions simultaneously:

1. M_z**-Sorting:** The states are arranged by the descending value of the total spin projection $M_z = \sum m_i$. This ensures that the Hamiltonian matrix has a block-diagonal structure.

2. **Spin-Inversion Centrosymmetry:** Within the M_z ordering, the sequence is chosen such that the entire basis is centrosymmetric with respect to the global spin inversion operator $\hat{P}$ (where $\hat{P}$ flips all spins $m_i \to -m_i$).

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2.2 Mathematical Definition of the Symmetry

This centrosymmetric property implies a rigorous mapping between the physical operation of spin inversion and the index numbers of the basis functions. If the dimension of the Hilbert space is $D = 2^N$, then for any k-th basis function:

$$\hat{P}|\psi_k\rangle = |\psi_{D+1-k}\rangle. \tag{5}$$

This relationship links the algebraic properties of the spin operators directly to the persymmetry (symmetry with respect to the anti-diagonal) of the Hamiltonian matrix.

3 Zeeman Interaction: Anti-Persymmetry

In the product basis, the operator $\hat{H}_Z$ is strictly diagonal since the basis functions are eigenfunctions of $\hat{I}_{iz}$.

3.1 Matrix Element

The energy of state $|\mathbf{a}\rangle$ is the sum of resonance frequencies weighted by spin projections:

$$\langle \mathbf{a}|\hat{H}_Z|\mathbf{a}\rangle = \sum_{i=1}^{N} \nu_i m_i^{(a)}. \tag{6}$$

3.2 Symmetry Property

Applying the inversion operator $\hat{P}$ flips all signs $m_i \to -m_i$, reversing the sign of the total Zeeman energy. Due to the centrosymmetric basis property ($k \leftrightarrow D+1-k$), this implies that the Zeeman matrix is antisymmetric with respect to the anti-diagonal (anti-persymmetric):

$$(H_Z)_{ii} = -(H_Z)_{D+1-i,D+1-i}. \tag{7}$$

4 Spin-Spin Interaction: Bisymmetry

4.1 General Matrix Element Formula

The full matrix element between states $|\mathbf{a}\rangle$ and $|\mathbf{b}\rangle$ is given by explicitly summing over all pairs:

$$\langle \mathbf{a}|\hat{H}_{SS}|\mathbf{b}\rangle = \sum_{k<l}^{N} \left[\underbrace{\delta_{\mathbf{a},\mathbf{b}}(J_{kl} + 2d_{kl}) m_k^{(a)} m_l^{(a)}}_{\text{Diagonal}} + \underbrace{\frac{1}{2}(J_{kl} - d_{kl})\hat{\Delta}_{kl}^{(\mathbf{a},\mathbf{b})}}_{\text{Off-diagonal}} \right]. \tag{8}$$

Where the auxiliary symbols are defined as follows:

- $\delta_{\mathbf{a},\mathbf{b}}$ (**Small Delta**) is the Kronecker delta, which selects the diagonal elements:

$$\delta_{\mathbf{a},\mathbf{b}} = \begin{cases} 1, & \text{if } |\mathbf{a}\rangle = |\mathbf{b}\rangle \\ 0, & \text{if } |\mathbf{a}\rangle \neq |\mathbf{b}\rangle \end{cases}. \tag{9}$$

- $\hat{\Delta}_{kl}^{(\mathbf{a},\mathbf{b})}$ (**Large Delta**) is the selection filter for flip-flop transitions. It equals 1 if and only if the states differ exactly by the exchange of spins k and l (where $m_k \neq m_l$):

$$\hat{\Delta}_{kl}^{(\mathbf{a},\mathbf{b})} = \begin{cases} 1, & \text{if } |\mathbf{b}\rangle \in \{\hat{I}_k^+ \hat{I}_l^- |\mathbf{a}\rangle,\ \hat{I}_k^- \hat{I}_l^+ |\mathbf{a}\rangle\} \\ 0, & \text{otherwise} \end{cases}. \tag{10}$$

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Note: While the secular Hamiltonian includes both scalar (J) and residual dipolar (d) couplings with different coefficients (see Eq. 8), both tensors share the same spatial symmetry properties defined by the molecular geometry. For brevity, in the subsequent symmetry analysis, we will refer to the generalized coupling matrix simply as the J-**coupling matrix**, implying that it effectively incorporates all spatial interaction constants.

4.2 Bisymmetry Property

The value of the spin-spin interaction matrix element is **invariant** under global spin inversion $\hat{P}$:

$$\langle \mathbf{a}|\hat{P}^\dagger \hat{H}_{SS}\hat{P}|\mathbf{b}\rangle = \langle \mathbf{a}|\hat{H}_{SS}|\mathbf{b}\rangle. \tag{11}$$

Conclusion: The $\hat{H}_{SS}$ matrix is symmetric with respect to both the main diagonal (Hermitian) and the anti-diagonal (persymmetric), making it a **bisymmetric matrix**. This property is universal.

5 Elementary Spectral Transformations

5.1 Spectrum Translation via Frequency Offset

In experimental NMR, changing the transmitter offset corresponds to a rigid translation of the spectrum.

5.1.1 Hamiltonian Transformation

Adding a constant offset Ω to all resonance frequencies ($\nu_i \to \nu_i + \Omega$) modifies the Hamiltonian:

$$\hat{H}' = \hat{H} + \Omega \hat{I}_z^{tot}. \tag{12}$$

Since $[\hat{H}, \hat{I}_z^{tot}] = 0$ in the secular approximation, the eigenstates remain unchanged, but the energy eigenvalues are shifted linearly: $E'_n = E_n + \Omega M_z^{(n)}$.

5.1.2 Shift of Spectral Lines

For observable single-quantum transitions ($\Delta M_z = -1$), the frequency shift is:

$$\omega'_{mn} = (E'_m - E'_n) = (E_m - E_n) + \Omega(M_z^{(m)} - M_z^{(n)}) = \omega_{mn} - \Omega. \tag{13}$$

Since the shift is identical for all lines, the internal multiplet structure (determined by J-couplings) remains invariant. Thus, the offset operation performs a pure translation of the spectral pattern.

5.2 Frequency Inversion and Spectral Reflection

It is a fundamental property that inverting the signs of all resonance frequencies leads to a mirror reflection of the spectrum. This reflection applies not only to the positions of the multiplets but also to their internal fine structure.

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5.2.1 Transformation of the Hamiltonian

Let $\hat{H}'$ be the Hamiltonian where all resonance frequencies are inverted ($\nu_i \to -\nu_i$). This is equivalent to:

$$\hat{H}' = -\hat{H}_Z + \hat{H}_{SS}. \tag{14}$$

Applying the spin inversion operator $\hat{P}$:

$$\hat{P}^\dagger \hat{H}' \hat{P} = -(-\hat{H}_Z) + \hat{H}_{SS} = \hat{H}_{original}. \tag{15}$$

Thus, $\hat{H}'$ and $\hat{H}_{original}$ are unitarily similar and share the **same set of energy eigenvalues**. This implies that the set of energy intervals (which define the magnitude of J-coupling splittings) remains numerically invariant.

5.2.2 Inversion of Selection Rules

Although the energy levels are the same, the observable transitions change due to the transformation of the eigenstates: $|\tilde{n}\rangle = \hat{P}|n\rangle$. The observable signal depends on the transition matrix elements of the lowering operator $\hat{I}^-$. The operator $\hat{P}$ transforms the lowering operator into the raising operator:

$$\hat{P}^\dagger \hat{I}^- \hat{P} = -\hat{I}^+. \tag{16}$$

Consequently, a transition allowed in the original system at frequency $\omega = E_f - E_i$ corresponds to a transition in the inverted system associated with the reverse process ($\Delta M = +1$), appearing at frequency $-\omega$.

5.2.3 Reflection of Fine Structure

Crucially, this mechanism reflects the **entire topology** of the spectrum. One might intuitively expect that since the scalar couplings J_{ij} are not inverted, the multiplets would simply shift positions while retaining their original shape. However, the unitary transformation $\hat{P}$ affects the mixing coefficients of the wavefunctions. As a result:

- The center of the multiplet moves from ν to $-\nu$.
- The internal structure is mirrored: a transition that was on the "right" side of the multiplet (e.g., $\nu + J/2$) maps to the "left" side relative to the new center ($-\nu - J/2$).
- Intensity distortions, such as the "roof effect" (where inner lines are stronger), are also strictly reflected.

Thus, $\hat{H}'(-\nu)$ generates a spectrum $S(-\omega)$ which is a perfect mirror image of $S(\omega)$ down to the finest detail.

5.3 Frequency Inversion and Magnet Reversal

Consider the transformation where all resonance frequencies are inverted ($\nu_i \to -\nu_i$), which is mathematically equivalent to $\hat{H}' = -\hat{H}_Z + \hat{H}_{SS}$. Physically, this corresponds to reversing the external magnetic field ($B_0 \to -B_0$).

- **Eigenvalues:** $\hat{H}'$ is unitarily equivalent to $\hat{H}$ via the spin inversion operator $\hat{P}$, so they share the same set of eigenvalues.
- **Selection Rules:** $\hat{P}$ transforms the lowering operator $\hat{I}^-$ into $\hat{I}^+$, effectively reversing the "direction" of transitions ($|n\rangle \to |m\rangle$ becomes $|\tilde{m}\rangle \to |\tilde{n}\rangle$).

Conclusion: The spectrum in a negative field is the mirror image of the spectrum in a positive field: $S(\omega) \to S(-\omega)$.

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6 Spectral Symmetry: The Ideal Case ($[\hat{H}, \hat{Q}] = 0$)

6.1 The Conflict of Symmetries

The total Hamiltonian $\hat{H} = \hat{H}_Z + \hat{H}_{SS}$ generally yields an asymmetric spectrum because its components transform differently under $\hat{P}$: $\hat{H}_Z$ is anti-symmetric, while $\hat{H}_{SS}$ is symmetric.

6.2 Restoration via Generalized Parity

Symmetry is restored if the "asymmetry" of the Zeeman term is compensated by the structural symmetry of the parameters. We introduce the **Generalized Parity Operator** $\hat{Q} = \hat{P} \times \hat{\Pi}$, where $\hat{\Pi}$ is the spatial permutation of nuclei ($i \leftrightarrow N + 1 - i$). The fundamental algebraic condition for symmetry is $[\hat{H}, \hat{Q}] = 0$.

6.3 Connection: Why Commutation Implies Symmetry

Why does the algebraic condition $[\hat{H}, \hat{Q}] = 0$ force the spectrum to be mirror-symmetric?

6.3.1 The Indistinguishability Argument

1. **Magnet Reversal Effect:** Changing the sign of the magnetic field reflects the spectrum: $S(\omega) \to S(-\omega)$ (Section 5.3).
2. **Structural Symmetry:** If the system parameters satisfy the conditions for $[\hat{H}, \hat{Q}] = 0$, then the state of the system in a reversed field ($-\boldsymbol{\nu}$) is structurally identical to the original state ($\boldsymbol{\nu}$), differing only by the labeling of the nuclei (operator $\hat{\Pi}$).
3. **Conclusion:** Since the physics is invariant under relabeling, the spectrum of the "reversed magnet" system must be identical to the spectrum of the "original" system:

$$S(-\omega) = S(\omega). \tag{17}$$

The commutation relation $[\hat{H}, \hat{Q}] = 0$ is the rigorous operator statement that the system is invariant under the combined operation of field inversion and relabeling.

6.4 The Generalized Canonical Basis ($\hat{Q}$-basis)

To make the spectral symmetry manifest in the matrix form, we introduce a reordered basis adapted to the combined operator $\hat{Q} = \hat{P}\hat{\Pi}$. This basis, which we term the **$\hat{Q}$-basis**, ensures that the pairing of states is defined by generalized parity, where each basis function is paired with its structural mirror image.

Mechanism of Frequency Restoration In a centered spectrum ($\sum \nu_i = 0$), the resonance frequencies obey the condition $\nu_k = -\nu_{N+1-k}$. Since the paired states in the $\hat{Q}$-basis are structural mirror images of each other, every individual spin contribution $+\nu_i$ in one state is perfectly compensated by a contribution $-\nu_i$ in its partner. This combined operation guarantees that if the resonance frequencies are antisymmetric relative to the center, the paired states possess identical Zeeman energies ($E = E'$).

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6.4.1 Mechanism of Symmetrization (Example)

Consider a 3-spin system. The state $|+--\rangle$ $(m_1 = +, m_2 = -, m_3 = -)$ transforms into its symmetric partner as follows:

$$|+--\rangle \xrightarrow{\hat{\Pi}} |--+\rangle \xrightarrow{\hat{P}} |++-\rangle.$$

This combined operation guarantees that if the resonance frequencies are antisymmetric relative to the center, the paired states possess identical Zeeman energies ($E = E'$). This transforms the Zeeman matrix for these **Shell** states from anti-persymmetric to **persymmetric**, matching the symmetry of the spin-spin interaction.

6.4.2 Structure and Symmetry of the Zero-Quantum Block ($M_z = 0$)

For subspaces with $M_z \neq 0$, the operator $\hat{Q}$ maps the entire block $+M$ to the block $-M$. However, in systems with an even number of spins, the Zero-Quantum subspace ($M_z = 0$) is mapped onto itself. It is only within this block that **self-conjugate states** (states invariant under $\hat{Q}$) can arise. To manage this complexity, the basis is organized into a **Nested "Shell-Core" Structure**, leading to a specific hybrid symmetry of the Zeeman interaction:

1. **The Shell (Transverse Pairs):** Consists of states that are *not* invariant under $\hat{Q}$ and form distinct pairs $|\psi_a\rangle \leftrightarrow |\psi_b\rangle$.
   - *Symmetry Implication:* Due to the frequency restoration mechanism, the Zeeman energies of the pair are equal ($E_a = E_b$). This equality transforms the Zeeman matrix representation from anti-persymmetry to **persymmetric**, aligning it with the symmetry of the spin-spin interaction.
2. **The Core (Self-Conjugate States):** Consists of states located in the exact center of the $M_z = 0$ block that are mapped onto themselves by $\hat{Q}$.
   - *Mechanism of Persistence:* For these states, the pairing is defined effectively by spin inversion $\hat{P}$, which reverses all spins without nuclear permutation, flipping the sign of the energy: $E \to -E$.
   - *Symmetry Implication:* Since there is no distinct partner with equal energy to restore balance, the condition $E_i = -E_j$ persists. Consequently, the Zeeman matrix in the Core remains **anti-persymmetric**.
   - *Algebraic Consistency:* This local violation of geometric bisymmetry is an artifact of the matrix representation for self-conjugate states and does not disrupt the fundamental commutation relation $[\hat{H}, \hat{Q}] = 0$. For such states, $\hat{Q}$ acts via sign inversion of Zeeman terms rather than by swapping degenerate partners, thus preserving algebraic symmetry.

Conclusion: The total Zeeman matrix in the $\hat{Q}$-basis possesses a **hybrid symmetry**: it is persymmetric in the outer shell but anti-persymmetric in the inner core. This prevents the total Hamiltonian $\hat{H} = \hat{H}_Z + \hat{H}_{SS}$ from being globally bisymmetric, while strictly maintaining the spectral palindromic property through its algebraic structure.

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6.5 Rigorous Mathematical Proofs

6.5.1 Proof via Unitary Transformation

The NMR spectrum $S(\omega)$ is the Fourier transform of the correlation function. Applying the unitary transformation $\hat{Q}$ to the trace (noting that $\hat{Q}^\dagger \hat{I}_x \hat{Q} = -\hat{I}_x$ and the trace is invariant under unitary similarity $\mathrm{Tr}(A) = \mathrm{Tr}(\hat{Q}^\dagger A \hat{Q})$):

$$\mathrm{Corr}'(t) = \mathrm{Tr}\left(e^{-i\hat{H}t}(-\hat{I}_x)e^{i\hat{H}t}(-\hat{I}_x)\right) = \mathrm{Corr}(t). \tag{18}$$

Since $\hat{Q}$ involves spin inversion, it effectively maps $S(\omega) \to S(-\omega)$. If the correlation function is invariant, the spectrum is symmetric.

6.5.2 Proof via Method of Moments

The symmetry can also be proven by showing that all odd Van Vleck moments vanish. The n-th moment M_n is defined as:

$$M_n = \int_{-\infty}^{\infty} \omega^n I(\omega) d\omega \propto \mathrm{Tr}\{[\hat{H}, [\hat{H}, \ldots [\hat{H}, \hat{I}_x] \ldots]]\hat{I}_x\}. \tag{19}$$

Applying the transformation $\hat{P}$ maps $\hat{H} \to \hat{H}'(-\nu)$. Since the spectrum of the inverted system is the mirror image of the original, $I(\omega) = I(-\omega)$. Consequently, all odd moments $M_1, M_3, \ldots$ must be identically zero.

6.6 Generalized Symmetry (Reflect and Shift)

If the system possesses structural symmetry (J_{ij} are centrosymmetric) but the transmitter offset is not set to zero (i.e., $\nu_i \neq -\nu_{N+1-i}$), the spectrum is still a palindrome, but centered at $\omega_0 \neq 0$. The generalized symmetry condition is:

$$S(2\omega_0 - \omega) = S(\omega). \tag{20}$$

This implies that the mirror image of the spectrum can be superimposed onto the original spectrum by a rigid linear translation.

7 The Case of $AA'BB'$: Proof of Symmetry via Symmetrized Basis

In C_2-symmetric spin systems of the $AA'BB'$ type, the balance of resonance frequencies is strictly ensured. Consequently, the matrix representation of the Hamiltonian in the $\hat{Q}$-ordered Product Basis corresponds to a palindromic structure. However, at the same time, this representation is **not invariant** under the action of the generalized parity operator $\hat{Q}$. This lack of invariance arises specifically from the inequality of the coupling constants ($J_{AA'} \neq J_{BB'}$), as a consequence of the fact that the matrix of J-coupling constants lacks symmetry with respect to the secondary diagonal. The action of the operator $\hat{Q}$ leads to the substitution of constants from the J-coupling matrix mirror-reflected relative to the secondary diagonal.

Nevertheless, it is important to note that from the explicit algebraic solution for the $AA'BB'$ system, it is known that the spectral energies are determined by the moduli of the sum and difference of the coupling constants $J_{AA'}$ and $J_{BB'}$. Consequently, the permutation of this pair of constants does not alter the spectrum, despite the explicit changes in the matrix representation. This implies that these constants form a **balanced pair**.

To clearly demonstrate the preservation of fundamental commutation relations despite apparent matrix differences, let us transform from the $\hat{Q}$-ordered Product Basis to the **C_2-Symmetrized Basis**.

7.1 Transition to the Symmetrized Basis

The C_2-symmetry operation involves the simultaneous interchange of nuclei indices corresponding to the chemically equivalent pairs ($A \leftrightarrow A'$ and $B \leftrightarrow B'$). Consequently, the total Hilbert space decomposes into two orthogonal subspaces corresponding to the irreducible representations of the symmetry group C_2:

1. **Symmetric Subspace ($\mathcal{H}_S$):** Spanned by linear combinations of product states that remain invariant under the interchange of primed and unprimed nuclei.
2. **Antisymmetric Subspace ($\mathcal{H}_A$):** Spanned by combinations that change sign under this interchange.

In this representation, the Hamiltonian matrix acquires a strict block-diagonal structure, eliminating the mixing between different symmetry manifolds:

$$\hat{H} = \hat{H}_S \oplus \hat{H}_A = \begin{pmatrix} \mathbf{H}_S & 0 \\ 0 & \mathbf{H}_A \end{pmatrix}. \tag{21}$$

This separation allows us to analyze the interaction of the Hamiltonian with the generalized parity operator $\hat{Q}$ independently for each subspace.

7.2 The Symmetric Block ($\hat{H}_S$): Strict Invariance

In the symmetric subspace, matrix elements depend exclusively on the sums of the coupling constants ($J_{AA'}$ and $J_{BB'}$); therefore, this subspace is invariant under the exchange $J_{AA'} \leftrightarrow J_{BB'}$. Since such sums remain unchanged by the permutation of parameters, this block strictly commutes with the parity operator:

$$[\hat{H}_S, \hat{Q}] = 0. \tag{22}$$

Thus, the subspectrum generated by $\hat{H}_S$ is a perfect palindrome.

7.3 The Antisymmetric Block ($\hat{H}_A$): Algebraic Isospectrality

In the antisymmetric subspace, the matrix elements depend on the **differences** of the coupling constants. To demonstrate the spectral symmetry of this block, we decompose the Hamiltonian $\hat{H}_A$ into four structural components based on their behavior under the generalized parity operator $\hat{Q}$:

$$\hat{H}_A = \underbrace{\hat{H}_Z + \hat{H}_{mean} + \hat{V}_{ext}}_{\hat{C}} + \underbrace{\hat{V}_{int}}_{\hat{V}}. \tag{23}$$

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7.3.1 Components of the Invariant Center ($\hat{C}$)

The component $\hat{C}$ consists of terms that are invariant under the action of the parity operator ($[\hat{C}, \hat{Q}] = 0$). It includes:

- **Zeeman Interaction ($\hat{H}_Z$):** Contains the frequency offsets (Δ) of the nuclei.
- **Mean Coupling Shift ($\hat{H}_{mean}$):** A global energy shift proportional to $-(J_{AA'} + J_{BB'})/2$, which does not affect the relative spectrum.
- **External Coupling Difference ($\hat{V}_{ext}$):** Proportional to $(J_{AB} - J_{AB'})$. Since this difference is invariant under $\hat{Q}$ in C_2 systems, it belongs to the invariant center.

7.3.2 The Symmetry-Breaking Term ($\hat{V}$)

The explicit non-invariance of the matrix representation is localized entirely within the internal coupling asymmetry:

- **Internal Coupling Asymmetry ($\hat{V}_{int}$):** This term is proportional to $(J_{AA'} - J_{BB'})$. Under the operation $\hat{Q}$, this difference undergoes a sign inversion ($J_{AA'} - J_{BB'} \to J_{BB'} - J_{AA'}$), leading to the anti-commutation $\{\hat{V}, \hat{Q}\} = 0$.

7.3.3 Preservation of the Spectrum

The anti-commutation of $\hat{V}$ with $\hat{Q}$ implies that the square of this term (and all its even powers) commutes with the parity operator: $[\hat{V}^2, \hat{Q}] = 0$. Since the eigenvalues of the Hamiltonian $\hat{H}_A$ depend on the symmetry-breaking term $\hat{V}$ effectively through its invariant properties (the characteristic polynomial remains unchanged when $\hat{V}$ is replaced by $-\hat{V}$), the spectrum of the antisymmetric block remains strictly palindromic. Thus, $J_{AA'}$ and $J_{BB'}$ function as a **balanced pair**, preserving the mirror symmetry of the total NMR spectrum.

7.4 Comparison with Magnetic Equivalence (A_nB_n and A_2X_2)

It is important to distinguish the $AA'BB'$ case from systems with magnetic equivalence, such as A_nB_n or A_2X_2.

In A_nB_n systems, the intra-group couplings (J_{AA} and J_{BB}) do not affect the transition energies, as they effectively disappear from the relevant portions of the Hamiltonian. Consequently, the condition for spectral mirror symmetry is satisfied trivially, regardless of the relative values of these constants.

In contrast, in $AA'BB'$ systems, the internal couplings $J_{AA'}$ and $J_{BB'}$ actively contribute to the structure of the spectrum. In this case, symmetry is not a result of the parameters' absence but is preserved only due to the specific algebraic structure of the $\hat{Q}$-basis (the "balanced pair" property). As demonstrated in the following section, the observed mirror symmetry is maintained through the cancellation of signs in squared differences within the Hamiltonian's characteristic polynomial, making $AA'BB'$ a unique case of algebraic isospectrality.

7.5 Conclusion

The apparent "defect" (non-invariance) in the $AA'BB'$ matrix representation is mathematically resolved. While $\hat{H}_S$ satisfies geometric symmetry, $\hat{H}_A$ satisfies the algebraic condition of anti-commutation for its variable part. Both mechanisms guarantee that the set of eigenvalues is invariant under reflection, resulting in a strictly palindromic NMR spectrum.

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8 The General Theorem of Spectral Symmetry

Based on the preceding analysis, we can formulate the most general condition for the mirror symmetry of an NMR spectrum. We introduce the concept of **reflected frequencies** relative to the spectral center of gravity $\nu_0 = \frac{1}{n}\sum \nu_i$. The reflected frequency set $\boldsymbol{\nu}'$ is defined as $\nu_i' = 2\nu_0 - \nu_i$.

Theorem 1 (Condition for Mirror Symmetry)**.** *An NMR spectrum $S(\omega)$ is mirror-symmetric with respect to its center **if and only if** the Hamiltonian with the reflected resonance frequencies, $\hat{H}(\boldsymbol{\nu}')$, is unitarily similar to the original Hamiltonian $\hat{H}(\boldsymbol{\nu})$.*

$$\exists\, \hat{U} : \quad \hat{H}(2\nu_0 - \nu, J) = \hat{U}^\dagger \hat{H}(\nu, J)\hat{U}. \tag{24}$$

Where $\hat{U}$ is a unitary operator.

Corollary 1.1 (Simultaneous Structural Requirement)**.** *The condition of unitary similarity implies a strict topological constraint on the spin system. For a spectrum to be mirror-symmetric, there must exist a **single specific spin ordering** (labeling sequence $1, \ldots, N$) that satisfies two conditions simultaneously:*

1. ***Frequency Equilibration:*** *The sequence of resonance frequencies is centrosymmetric about the spectral center ν_0 (e.g., $\nu_i - \nu_0 = -(\nu_{N+1-i} - \nu_0)$).*
2. ***Matrix Reflection Invariance:*** *In this same ordering, the J-coupling matrix is invariant under reflection about the secondary diagonal. This invariance can be either:*
   - ***Explicit (Geometric):*** *$J_{ij} = J_{N+1-i,N+1-j}$ (balanced systems like A_nB_n).*
   - ***Implicit (Isospectral):*** *$J_{ij} \neq J_{N+1-i,N+1-j}$, but the pairs are "balanced" such that the spectrum is isospectral under their permutation (as in $AA'BB'$).*

If no such single ordering exists (i.e., the ordering required to center the frequencies disrupts the necessary J-coupling matrix symmetry, as in heteronuclear cycles like cyclooctatetraene derivatives), the spectral symmetry is broken.

9 Practical Criterion for Spectral Symmetry

Based on the general theorem, we can formulate a practical criterion for verifying the mirror symmetry of an NMR spectrum without full diagonalization.

Criterion 1 (Necessary and Sufficient Conditions)**.** *The NMR spectrum $S(\omega)$ is mirror-symmetric with respect to the central frequency ν_0 **if and only if** there exists a spin ordering index $k \in \{1, \ldots, N\}$, sorted by resonance frequencies, for which the following two conditions hold simultaneously:*

Condition 1: Frequency Centrosymmetry *The distribution of resonance frequencies must be antisymmetric with respect to the spectral center ν_0. For every nucleus k:*

$$\nu_k - \nu_0 = -(\nu_{N+1-k} - \nu_0). \tag{25}$$

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Condition 2: Symmetry of the J-Coupling Matrix *Consider the operation of reflection about the secondary diagonal, denoted by $\mathcal{R}$, acting on the J-coupling matrix $\boldsymbol{J}$:*

$$(\mathcal{R}\boldsymbol{J})_{ij} = J_{N+1-i,\,N+1-j}. \tag{26}$$

For mirror symmetry to exist, the system must satisfy one of the following two cases:

- ***Case A: Geometric Symmetry (Explicit)***
  The J-coupling matrix is invariant under reflection:

  $$\boldsymbol{J} = \mathcal{R}\boldsymbol{J} \quad \Longleftrightarrow \quad J_{ij} = J_{N+1-i,\,N+1-j}. \tag{27}$$

  This corresponds to systems with physical geometric symmetry (e.g., AB or AX).

- ***Case B: Isospectral Symmetry of Balanced Pairs (Implicit)***
  *The matrix changes under reflection ($\boldsymbol{J} \neq \mathcal{R}\boldsymbol{J}$), but the spectrum remains invariant. This occurs when the non-matching coupling constants form **balanced pairs** $\{J_\alpha, J_\beta\}$ that are swapped by the reflection operation:*

  $$J_\alpha \xrightarrow{\mathcal{R}} J_\beta, \quad J_\beta \xrightarrow{\mathcal{R}} J_\alpha \quad (\textit{where } J_\alpha \neq J_\beta). \tag{28}$$

  Invariance Requirement: *The set of energy eigenvalues must be invariant under the permutation of constants within these pairs:*

  $$\textit{Spect}(\dots, J_\alpha, J_\beta, \dots) = \textit{Spect}(\dots, J_\beta, J_\alpha, \dots). \tag{29}$$

  Physical interpretation: This implies that the characteristic polynomial of the Hamiltonian depends on these parameters in a symmetric fashion (e.g., involving only their sum $J_\alpha + J_\beta$ and the square of their difference $(J_\alpha - J_\beta)^2$), as observed in $AA'BB'$ systems.

*In systems containing large groups of chemically equivalent nuclei, e.g., $[A'B']_n$, the requirement of sorting by resonance frequencies is necessary but not sufficient. Due to the permutation symmetry within equivalent groups, there exist multiple spin orderings that satisfy the frequency sorting condition. However, as matrix analysis demonstrates, the condition of J-coupling matrix symmetry is sensitive to the relative ordering of spins within these chemically equivalent groups. Therefore, for such high-order systems, the criterion is strictly defined as follows: Mirror symmetry exists if there is **at least one specific permutation** within the chemically equivalent sets that results in a symmetric J-coupling matrix.*

10 Implications for Structure Elucidation (The Inverse Problem)

Since the criterion established above provides necessary and sufficient conditions, the experimental observation of a mirror-symmetric spectrum imposes strict constraints on the properties of the underlying Hamiltonian. This allows for the partial solution of the "inverse problem": inferring structural properties solely from spectral symmetry.

1. **Unitary Self-Similarity:** A Hamiltonian generating a symmetric spectrum is guaranteed to be unitarily similar to itself under the operation of frequency reflection. Mathematically, there exists a unitary operator $\hat{U}$ such that:

   $$\hat{H}(2\nu_0 - \nu) = \hat{U}^\dagger \hat{H}(\nu) \hat{U}. \tag{30}$$

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2. **Topological Constraints:** The molecular structure must admit at least one numbering of nuclei where the J-coupling matrix $\boldsymbol{J}$ exhibits either explicit geometric symmetry (reflection invariance) or implicit isospectral symmetry (balanced pairs).

3. **Exclusion of Candidates:** In practical structure elucidation, the presence of a symmetric multiplet allows one to immediately reject any candidate structures (e.g., arbitrary $ABCD$ spin systems with random couplings) that do not possess the required topological symmetry in their J-coupling network.

11 Conclusion

The mirror symmetry of NMR spectra is governed by the interplay between the distribution of resonance frequencies and the topology of the J-coupling network. We have established that:

1. **Necessary Condition:** The Hamiltonian must be unitarily similar to itself under the operation of frequency reflection relative to the spectral center.

2. **Practical Criterion:** This similarity is physically realized if and only if there exists a spin permutation that **simultaneously**:
   - Arranges resonance frequencies symmetrically about the center.
   - Reveals a J-coupling matrix structure that is invariant (geometrically or isospectrally) under reflection about the secondary diagonal.

Systems with full magnetic equivalence (such as A_nB_n or A_nX_n) satisfy these conditions trivially because the symmetry-breaking intra-group couplings do not affect the spectrum. Systems like $AA'BB'$ satisfy them via algebraic isospectrality (balanced pairs) **originating from the internal symmetry of the spin system**. However, in more complex systems where chemical equivalence exists but magnetic equivalence is violated (general structures of type $[A'B']_n$ or complex heteronuclear cycles), the specific topological balance required for isospectrality may be absent. In such cases, the conflict between frequency equilibration and J-coupling matrix symmetry results in **asymmetric spectra**.

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